

6061-T6/T651 Aluminum Alloy

Data Sheet

Basic Information

Material Grade: Aluminum Alloy 6061-T6; 6061-T651

Subcategories: [6000 Series Aluminum Alloy](#); Aluminum Alloy; Metal; Non-Ferrous Metal

Keywords: al6061, UNS A96061; ISO Aluminum-Magnesium-Silicon-Copper; Aluminum 6061-T6, AD-33 (Russia); AA6061-T6; 6061T6, UNS A96061; ISO Aluminum-Magnesium-Silicon-Copper; [Aluminum 6061-T651](#), AD-33 (Russia); AA6061-T651

Composition Description

The listed aluminum content is a balance value, meaning the remaining portion after deducting the weight percentages of all other alloying elements is aluminum. This composition information is provided by the Aluminum Association for reference only and **is not suitable for engineering design, product development, or other precision scenarios**. The content range of all alloying elements complies with the general standards for 6061 series aluminum alloys, ensuring the material possesses corresponding mechanical and physical-chemical properties.

Application Fields

Aircraft parts, camera lens mounts, couplings, marine fittings and hardware, electrical fittings and connectors, decorative or miscellaneous hardware, hinge pins, magneto parts, brake pistons, hydraulic pistons, electrical accessories, valves and valve parts; bicycle frames.

Physical Properties

Property Index	Value (Metric)	Value (Imperial)	Remarks
Density	2.7 g/cm ³	0.0975 lb/in ³	AA; Typical Value

Mechanical Properties

Property Index	Value (Metric)	Value (Imperial)	Remarks
Brinell Hardness	95	95	AA; Typical Value; 500g Load; 10mm Ball
Knoop Hardness	120	120	Converted from Brinell Hardness value
Rockwell A Hardness	40	40	Converted from Brinell Hardness value
Rockwell B Hardness	60	60	Converted from Brinell Hardness value
Vickers Hardness	107	107	Converted from Brinell Hardness value
Ultimate Tensile Strength	310 MPa	45,000 psi	AA; Typical Value
Tensile Yield Strength	276 MPa	40,000 psi	AA; Typical Value
Elongation at Break	12%	12%	AA; Typical Value; Thickness 1/16 inch (1.6 mm)
Elongation at Break	17%	17%	AA; Typical Value; Diameter 1/2 inch (12.7 mm)
Modulus of Elasticity	68.9 GPa	10,000 ksi	AA; Typical Value; Average of tension and compression. Compressive modulus is approximately 2% higher than tensile

			modulus.
Notched Tensile Strength	324 MPa	47,000 psi	Side-notched specimen, 2.5 cm wide × 0.16 cm thick, Kt = 17
Ultimate Bearing Strength	607 MPa	88,000 psi	Edge Distance/Pin Diameter = 2.0
Bearing Yield Strength	386 MPa	56,000 psi	Edge Distance/Pin Diameter = 2.0
Poisson's Ratio	0.33	0.33	Estimated based on trends of similar aluminum alloys
Fatigue Strength	96.5 MPa	14,000 psi	AA; Fully reversed stress, 500 million cycles; RR Moore machine/specimen
Fracture Toughness	29 MPa·m ^{1/2}	26.4 ksi·in ^{1/2}	KIC; TL Orientation
Machinability	50%	50%	Aluminum Alloy Scale 0-100
Shear Modulus	26 GPa	3,770 ksi	Estimated based on similar aluminum alloys
Shear Strength	207 MPa	30,000 psi	AA; Typical Value

Electrical Properties

Property Index	Value	Remarks
Electrical Resistivity	3.99e-006 Ω·cm	AA; Typical Value at 68°F

Thermal Properties

Property Index	Value (Metric)	Value (Imperial)	Remarks
Linear CTE (68°F)	23.6 $\mu\text{m}/\text{m}\cdot^{\circ}\text{C}$	13.1 $\mu\text{in}/\text{in}\cdot^{\circ}\text{F}$	AA; Typical Value; Average in the range of 68-212°F
Linear CTE (250°C)	25.2 $\mu\text{m}/\text{m}\cdot^{\circ}\text{C}$	14 $\mu\text{in}/\text{in}\cdot^{\circ}\text{F}$	Estimated based on trends of similar aluminum alloys; 20-300°C
Specific Heat Capacity	0.896 J/g·°C	0.214 BTU/lb·°F	-
Thermal Conductivity	167 W/m·K	1160 BTU-in/hr-ft²·°F	AA; Typical Value at 77°F
Melting Point	582 - 652 °C	1080 - 1205 °F	AA; Typical range based on typical composition of forged products with thickness 1/4 inch or more; Eutectic melting can be completely eliminated by homogenization.
Solidus Temperature	582°C	1080 °F	AA; Typical Value
Liquidus Temperature	652 °C	1205 °F	AA; Typical Value

Processing Characteristics

Process Parameter	Temperature (Metric)	Temperature (Imperial)	Remarks
Solution Treatment	529°C	985°F	-

Temperature			
Aging Temperature (Rolled/Stretched Products)	160°C	320 °F	Hold for 18 hours
Aging Temperature (Extruded/Forged Products)	177°C	350 °F	Hold for 8 hours

Notes

To maintain consistency in information format, some of the values shown above may have been converted from original units and/or rounded. Users requiring more precise data for scientific or engineering calculations can click on the property values to view the original values and their corresponding converted original units. We recommend using only the original values or their converted original units in calculations to minimize rounding errors.